

The target represented on the CAR trace GNS space

MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation

def

A fixed pointed unitary transports the target action onto the Hilbert space constructed from CAR alone.

Statement

Let A be the completed CAR algebra and τ its normalized trace. Fix a representative shell family and its chosen unitary links L_j in the selected atomic representation π . Write $T = C^*(\pi(A), \{L_j\})$ for the resulting concrete target and $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, for its faithful unital source map. This fixes one target and one source map throughout. Let $(H_\tau, \pi_\tau, \xi_\tau)$ be the GNS representation of the CAR trace, and let $(H_\phi, \rho_\phi, \xi_\phi)$ be the target-state GNS representation for the chosen extension ϕ . Fix the pointed unitary $E : H_\tau \rightarrow H_\phi$ described below. Define the target representation on H_τ by

$$\tilde{\rho}(t) = E^{-1} \rho_\phi(t) E \quad (t \in T).$$

Definition

Set $H_\tau = L^2(A, \tau)$, $\xi_\tau = [1]$, and $\pi_\tau(a)[b] = [ab]$. The GNS identities give $\|\xi_\tau\| = 1$ and $\tau(a) = \langle \xi_\tau, \pi_\tau(a) \xi_\tau \rangle$. Thus $H_\tau \neq 0$, and its dense continuous CAR orbit makes it separable. The target GNS has the same source vector functional and a dense source orbit. Pointed cyclic transport therefore gives, and `traceGNSUnitary` chooses once,

$$E\xi_\tau = \xi_\phi, \quad E\pi_\tau(a) = \rho_\phi(\iota(a))E.$$

Conjugate the whole target action by its inverse. Then $E^{-1} \rho_\phi(\iota(a)) E = \pi_\tau(a)$ and

$$\langle \xi_\tau, \tilde{\rho}(t) \xi_\tau \rangle = \langle E\xi_\tau, \rho_\phi(t) E\xi_\tau \rangle = \phi(t).$$

The dense $\pi_\tau(A)\xi_\tau$ orbit is contained in the target orbit, proving target cyclicity. The defining RHS below uses the inverse of this same chosen E .

Assumptions

The representative shell family and chosen state extension are fixed. Both GNS representations restrict to the same source trace, and both source orbits are dense; source cyclicity in H_ϕ is a proved result.

Conclusion

The new representation has $\tilde{\rho}(\iota(a)) = \pi_\tau(a)$ and realizes ϕ at ξ_τ . Its target orbit is dense. The space H_τ is nonzero and separable and is defined from CAR alone, independently of the family; the target action still depends on the fixed family.

The direction $E : H_\tau \rightarrow H_\phi$ determines the conjugation $E^{-1}\rho_\phi E$. This is one chosen pointed unitary, used throughout the definition and its consequences. Injectivity of this representation is stated separately. The target T itself has not been replaced.

Main citations

- Definition and its exact construction — `MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation`
- The fixed concrete target and source map —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`
- Faithfulness of the source embedding —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom_injective`
- CAR-only trace Hilbert space — `MathlibAnnex.CStarAlgebra.CAR.TraceHilbertSpace`
- CAR trace cyclic vector — `MathlibAnnex.CStarAlgebra.CAR.traceVector`
- CAR trace GNS action — `MathlibAnnex.CStarAlgebra.CAR.traceRepresentation`
- Unit norm of the CAR vector — `MathlibAnnex.CStarAlgebra.CAR.norm_traceVector`
- Nonzero CAR vector — `MathlibAnnex.CStarAlgebra.CAR.traceVector_ne_zero`
- Nonzero CAR trace Hilbert space —
`MathlibAnnex.CStarAlgebra.CAR.nontrivial_traceHilbertSpace`
- CAR vector-state identity —
`MathlibAnnex.CStarAlgebra.CAR.inner_traceVector_traceRepresentation`
- Dense CAR trace orbit —
`MathlibAnnex.CStarAlgebra.CAR.denseRange_traceRepresentation_orbit`
- Separable CAR trace Hilbert space —
`MathlibAnnex.CStarAlgebra.CAR.separableSpace_traceHilbertSpace`
- Existence of the pointed source unitary —
`MathlibAnnex.CStarAlgebra.CAR.exists_linearIsometryEquiv_traceHilbertSpace`
- The unitary chosen once — `MathlibAnnex.CStarAlgebra.CAR.traceGNSUnitary`
- Source cyclicity of the target-state GNS —
`MathlibAnnex.CStarAlgebra.CAR.denseRange_tracialRepresentation_source_orbit`
- Restriction of the transported target action —
`MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation_shellFamilySourceHom`
- Transported vector-state identity —
`MathlibAnnex.CStarAlgebra.CAR.inner_traceVector_traceModelRepresentation`
- Cyclicity of the transported target action —
`MathlibAnnex.CStarAlgebra.CAR.denseRange_traceModelRepresentation_orbit`

Supporting route explanation

Lean source signature (exact)

```
noncomputable def traceModelRepresentation (family : RepresentativeShellFamily) :  
  Representation (ShellFamilyTarget family) TraceHilbertSpace :=
```

```
(traceGNSUnitary family).symm.conjStarAlgEquiv.toStarAlgHom.comp
(tracialRepresentation family)
```

In the source	Mathematical meaning
<code>family : RepresentativeShellFamily;</code> <code>ShellFamilyTarget family</code>	The fixed shell family and its unchanged target T , source ι , and chosen trace extension ϕ .
<code>TraceHilbertSpace</code>	The complete CAR trace GNS space $H_\tau = L^2(A, \tau)$, with $\pi_\tau(a)[b] = [ab]$ and $\xi_\tau = [1]$. This is different from the target-state GNS space H_ϕ .
<code>traceGNSUnitary family</code>	The once-chosen pointed unitary $E : H_\tau \rightarrow H_\phi$, satisfying $E\xi_\tau = \xi_\phi$ and $E\pi_\tau(a) = \rho_\phi(\iota(a))E$.
<code>(traceGNSUnitary</code> <code>family).symm.conjStarAlgEquiv.toStarAlgHom</code>	The same inverse unitary sends an operator R on H_ϕ to $E^{-1}RE$ on H_τ ; the direction is determined by <code>.symm</code> .
<code>... .comp (tracialRepresentation family)</code>	The complete RHS composes that operator conjugation with $t \mapsto \rho_\phi(t)$, yielding $\tilde{\rho}(t) = E^{-1}\rho_\phi(t)E$ for every $t \in T$.
<code>Representation (ShellFamilyTarget family)</code> <code>TraceHilbertSpace</code>	The output is the target representation $\tilde{\rho} : T \rightarrow B(H_\tau)$. It has not replaced T or identified the two GNS spaces without transport; injectivity is separately proved.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation`

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